



Final Project Data Analysis in Rehabilitation

Prof. Joan F. Alonso

Master in
Neuroengineering
and
Rehabilitation

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1 Introduction and Aim

1.1 Some context

Welcome to the integrative data analysis project! Over the next weeks, we will embark on a journey into the field of Brain-Computer Interfaces (BCIs). We will analyse real electroencephalography (EEG) data recorded from volunteers performing motor imagery tasks. Specifically, volunteers imagined moving either their left or their right hand, or they remained in a relaxed state (no movement). We will start with raw data, navigate the challenges of biological signal processing, explore feature engineering techniques, and apply machine learning models to decode the brain's intentions. This project provides hands-on experience with a complete data analysis workflow in a challenging and exciting application domain. In the *Human-Machine Interfaces (HMI)* course we have already explored the decoding of upper-limb movements using the database described in [P. Ofner et al., 2017](#). Specifically, we now know how to:

- **Band-pass filter EEG signals** and identify bad channels and artefacts.
- **Select usable trials** using trimmed normalisation.
- **Re-reference** signals (common average, Laplacian, etc.).
- **Assess movement execution** on the EEG signals by event-related de/synchronisation (ERD/ERS) and slow cortical potentials (SCP).
- **Represent topographic maps** to check the channels where these potentials are best observed.

Therefore, in this project we will use this knowledge to decipher the movement imagined by a person.

1.2 Key points

- **Primary aim of the project**
To design, implement, and evaluate a machine learning pipeline capable of accurately classifying different motor imagery states (left-hand, right-hand imagery, no imagery) based solely on the recorded EEG signals.
- **Duration**
6 weeks (from 27th April to 7th June, 2026)
- **Group Size**
4 students
- **Grading**
70 % Report, 30 % Presentation.
Details on the deliverables are in [Section 3.2](#). The rubrics used to assess the project are available in [Annexe B](#) and [Annexe C](#).

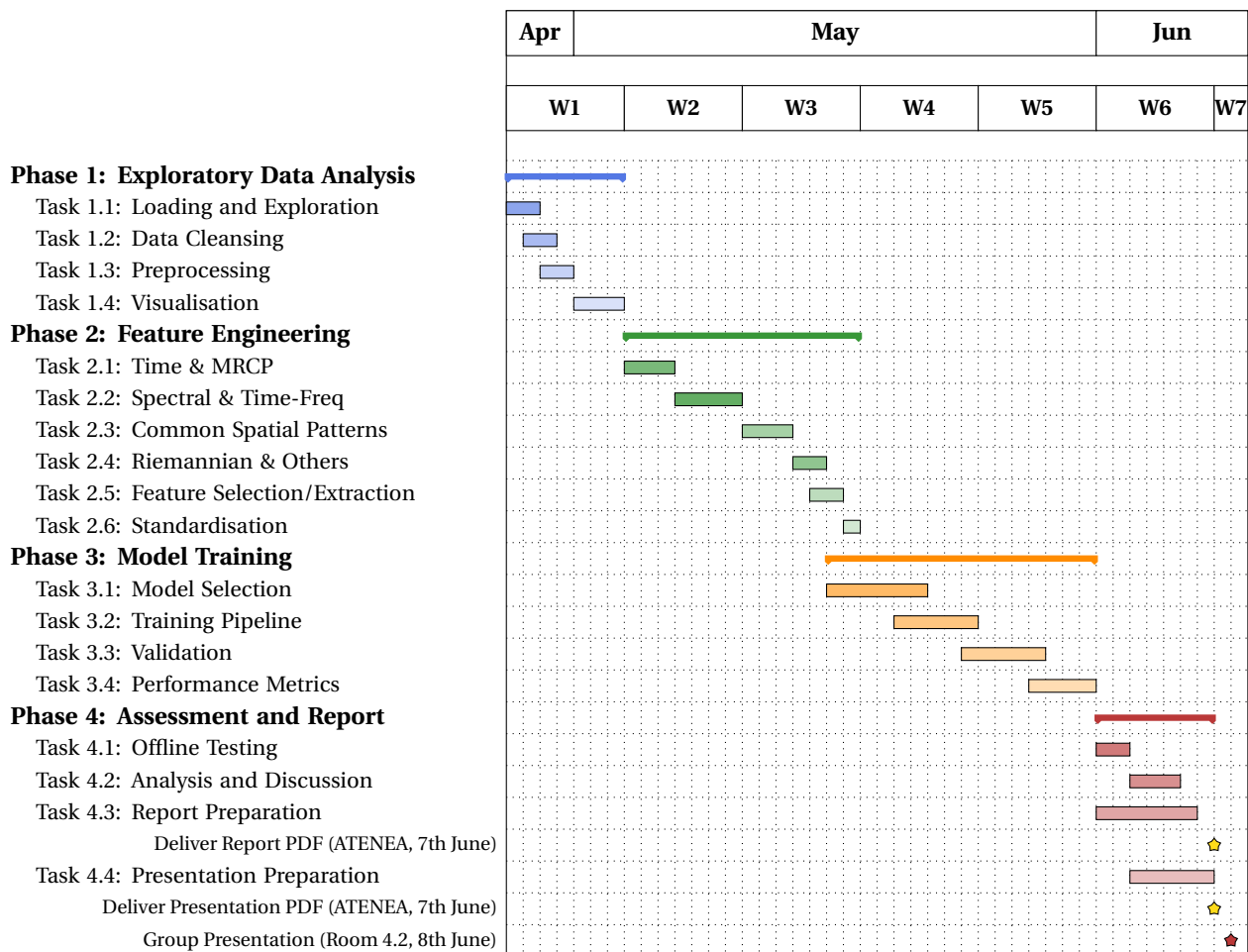
1.3 Reporting the use of AI tools

According to the current recommendations at the Universitat Politècnica de Catalunya (UPC), students must be transparent about the use of AI tools.

- You must include a section in your report declaring which AI tools were used (e.g., ChatGPT, Claude, GitHub Copilot).
- Clearly specify how they assisted your work (e.g., code debugging, text polishing, or brainstorming).
- Remember that you are responsible for the final content; any "hallucinations" or errors provided by the AI and included in the project will be penalised.

2 Project Phases and Tasks

2.1 Proposed Planning



2.2 Phase 1: Exploratory Data Analysis

- **Objective:** Familiarise yourselves with the EEG dataset and prepare it for analysis.
- **Tasks:**
 - **Data Loading and Exploration:** Load [the provided EEG datasets \(in EEGLab .mat format\)](#). Understand the data structure, channel layout (based on the 16-channel OpenBCI setup), and event markers indicating the different imagery tasks. [Annexe A](#) describes the contents and format of the shared files.
 - **Data Cleansing:** Identify and handle potential issues like noise, artefacts (e.g., blinks, ocular movements, muscular activity from the head and neck), and bad channels. Decide on appropriate strategies for correction and/or removal.
 - **Preprocessing:** Apply necessary preprocessing steps such as filtering (e.g., bandpass filtering to isolate relevant frequency bands), referencing, and epoching (segmenting the data into trials based on task markers).
 - **Data Visualisation:** Create informative visualisations (e.g., time series plots, power spectral density plots, topographic plots) to understand the characteristics of the EEG signals during different tasks and to guide preprocessing choices.
- **Comments:** It may be easier to do this in MATLAB, reusing code developed in *HMI*.

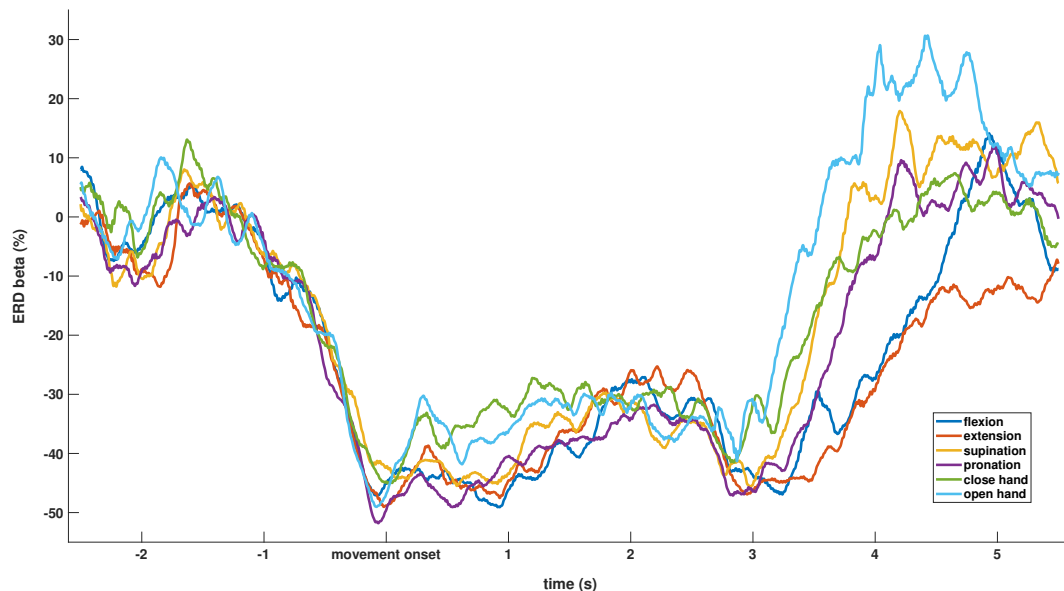


Figure 1: Grand-average ERD/ERS potentials obtained in the β band for the movements in [P. Ofner et al., 2017](#). Image by S. Romero.

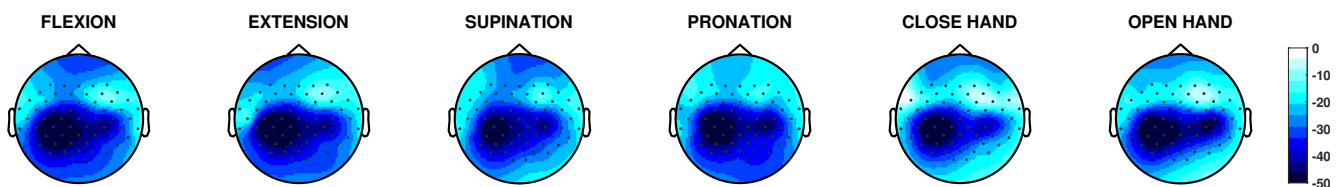


Figure 2: Grand-average ERD potentials (in μV) obtained in the β band at the movement onset, for the movements in [P. Ofner et al., 2017](#). Image by S. Romero.

2.3 Phase 2: Feature Engineering and Selection/Extraction

- **Objective:** Create meaningful features from the preprocessed EEG data that can aid discrimination between the different motor imagery states.
- **Tasks:**
 - **Brainstorming and Feature Engineering:** Explore and implement various feature types relevant to motor imagery EEG, including:
 - * **Time-domain features:** Explore Movement-Related Cortical Potentials (MRCPs), although these are often more prominent in actual movement, investigate if relevant components exist in the imagery data. These potentials are usually best detected when the signal is low-pass filtered (e.g. from 0.3 Hz to 3 Hz). Keep in mind that slow cortical potentials are best detected around central locations (Cz and neighbouring electrodes), as seen in the maps in [Figure 4](#).
 - * **Frequency-based features:** Investigate ERD/ERS patterns, typically observed in the μ (8 Hz to 13 Hz) and β (14 Hz to 30 Hz) bands over motor cortex areas during motor imagery (See [Figure 1](#) and [Figure 2](#)). Calculate band power or related metrics.
 - * **Time-Frequency Analysis:** Implement Wavelet Transforms or Short-Time Fourier Transforms (STFT) to visualise how frequency power shifts *during* the trial window rather than using static averages.
 - * **Common Spatial Patterns (CSP):** Implement and utilise CSP, a powerful technique for BCI that finds spatial filters maximising variance between classes. Experiment with the number of patterns/filters. Additional insights on CSP [in this article by B. Blankertz et al. 2007](#).
 - * **Riemannian Geometry & Other potential features:** As a modern alternative to CSP, explore covariance matrices projected into Tangent Space. This is often more robust to noise and smaller datasets. Additionally, investigate other statistical measures or connectivity metrics.

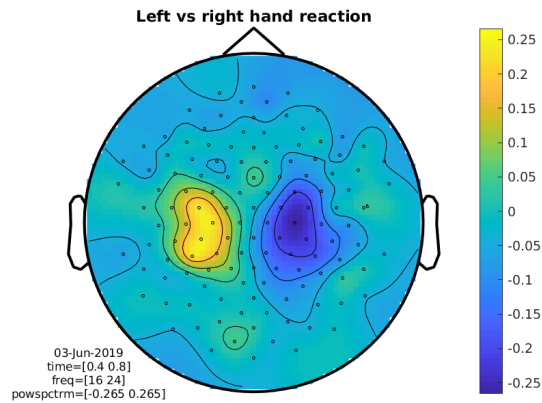


Figure 3: Topographic representation of the difference in beta power between left and right response. Taken from [a FieldTrip tutorial](#)

– **Feature Analysis and Selection/Extraction:**

- * Analyse the discriminative power of individual features or feature sets.
- * Use techniques like training and validation curves to assess potential overfitting or underfitting with different feature sets.
- * Evaluate feature redundancy (e.g., using correlation matrices).
- * Decide whether to use feature selection methods (choosing a subset of the best features) or feature extraction methods (like PCA or potentially using CSP itself as a dimensionality reduction technique). Justify your choices.

- **Standardisation/Normalisation:** Apply appropriate scaling techniques to ensure features are comparable.

• **Comments:**

- The articles by [D. J. McFarland et al., 2005](#) and by [S. Vaid et al., 2015](#) can assist in discovering other useful features in BCI systems.
- Feature engineering may also be easier in MATLAB, adapting *HMI* codes. CSP can also be calculated easily [using MATLAB](#). A Python alternative ([MNE-Python package](#)), should you wish to code everything in Python, is also very useful (see [example](#)).
- Below (see pages 5 to 6) you can find a basic example of:
 - * a code structure to process signals and calculate features;
 - * how to use a MATLAB table to store features;
 - * how to export a table to a .csv file.

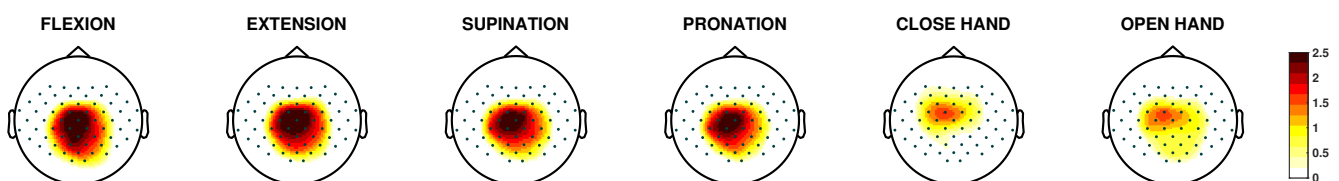


Figure 4: Slow cortical potential topographies for the movements in [P. Ofner et al., 2017](#). Maps show the grand-average amplitude in μV of a 0.5 s segment selected just before the movement onset. Image by S. Romero.

```

1 %% START & CLEANUP
2 clearvars; close all; clc;
3
4 %% LOAD EEG SIGNALS FROM A VOLUNTEER
5 filename = 'v01.mat';
6 load(filename); % Load EEG variable, EEGLab format
7
8 %% DEFINITIONS
9 num_events = numel(EEG.event); % 960 --> 480 pairs of fixation cross (c) followed by task
10 %           First 120 trials are motor execution
11 %           Next 360 are motor imagery. Each trial is labeled:
12 %           - b indicates resting (baseline)
13 %           - r indicates right
14 %           - l indicates left
15 Fs = EEG.srate; % 125 Hz
16 signals = detrend(EEG.data(1:15,:),'constant'); % 16 columns (but ch 16 unused)
17 % Ignore EEG labels in the mat-file. These are the channels in order:
18 ch_names = {'F7','F3','Fz','F4','F8','T3','C3','Cz','C4','T4','P7','P3','Pz','P4','P8'};
19 first_trial = 121;
20
21
22 %% FILTER DEFINITIONS
23 % Mu+Beta filter (8 to 30 Hz)
24 Fstop1 = 7.9; % First Stopband Frequency
25 Fpass1 = 8; % First Passband Frequency
26 Fpass2 = 30; % Second Passband Frequency
27 Fstop2 = 30.1; % Second Stopband Frequency
28 Astop1 = 20; % First Stopband Attenuation (dB)
29 Apass = 0.1; % Passband Ripple (dB)
30 Astop2 = 20; % Second Stopband Attenuation (dB)
31 match = 'passband'; % Band to match exactly
32 h = fdesign.bandpass(Fstop1, Fpass1, Fpass2, Fstop2, Astop1, Apass, Astop2, Fs);
33 mubeta_filter = design(h, 'cheby2', 'MatchExactly', match, 'SystemObject', true);
34 reorder(mubeta_filter, 'up');
35
36
37 %% EPOCHING TRIALS AND CONCATENATION
38 mi_right = [];
39 mi_left = [];
40 mi_rest = [];
41 mi_right_std = [];
42 mi_left_std = [];
43 mi_rest_std = [];
44
45 for ev = first_trial*2 : 2 : num_events
46     start_sample = EEG.event(ev).latency;
47     stop_sample = EEG.event(ev).latency+2*Fs-1;
48
49     % Display trial latency
50     fprintf('Ev %3d [%s]--> %6d (%8.3f)\n', ...
51         ev, upper(EEG.event(ev).type), start_sample, start_sample/Fs);
52
53     trial = detrend(signals(start_sample:stop_sample, :), 'constant');
54     pad_trial = [flipud(trial); trial; flipud(trial)];
55     filt_pad_trial = filtfilt(mubeta_filter.SOSMatrix, mubeta_filter.ScaleValues, pad_trial);
56     filt_trial = filt_pad_trial(size(trial,1)+1:2*size(trial,1), :);
57
58     % Plot to check that filtered signals look OK
59     if ev == first_trial*2
60         figure; hold on; box on;
61         plot(1/Fs:1/Fs:length(trial(:,1))/Fs, trial(:, strcmp(ch_names, 'Cz')));
62         plot(1/Fs:1/Fs:length(trial(:,1))/Fs, filt_trial(:, strcmp(ch_names, 'Cz')));
63         legend('Raw Cz (detrended)', 'Filtered Cz');
64         ylabel('amplitude (\muV)', 'Interpreter','tex');

```

```

65     xlabel('time (s)')
66     title(['EEG (' filename ')'])
67     axis tight;
68 end
69
70 % Calculations performed trial by trial:
71 % - Features (for example, standard deviation of channel Cz):
72 % - Concatenate trial to corresponding matrix (for example for later SCP calculation)
73 switch EEG.event(ev).type
74     case 'r'
75         mi_right_std = [mi_right_std; std(filt_trial(:,strcmp(ch_names, 'Cz')))];
76         mi_right = [mi_right; filt_trial];
77     case 'l'
78         mi_left_std = [mi_left_std; std(filt_trial(:,strcmp(ch_names, 'Cz')))];
79         mi_left = [mi_left; filt_trial];
80     case 'b'
81         mi_rest_std = [mi_rest_std; std(filt_trial(:,strcmp(ch_names, 'Cz')))];
82         mi_rest = [mi_rest; filt_trial];
83     otherwise
84         error('This should not happen!')
85 end
86 end
87
88 %% COMBINE VECTORS INTO TABLE & EXPORT CSV
89 % Use only 4 values of each task (instead of 120), just for the example:
90 df = table([mi_rest_std(1:4); mi_left_std(1:4); mi_right_std(1:4)] , ...
91           [repmat("rest",4,1); repmat("left",4,1); repmat("right",4,1)], ...
92           'VariableNames', {'std_8_30', 'mi_task'});
93 writetable(df, 'features.csv')

```

————— Exported 'features.csv' file —————

```

1  std_8_30,mi_task
2  0.6183329,rest
3  0.6678268,rest
4  0.5968277,rest
5  0.5260289,rest
6  0.6791871,left
7  0.7106078,left
8  0.5141629,left
9  0.7058661,left
10 0.7243008,right
11 0.6787799,right
12 0.6231062,right
13 0.596122,right

```

2.4 Phase 3: Model Training and Validation

- **Objective:** Train and rigorously evaluate different machine learning models for the classification task.
- **Tasks:**
 - **Model Selection:** Choose at least 2 or 3 different classification algorithms suitable for this type of data (e.g., Linear Discriminant Analysis (LDA), Logistic Regression (LR), k-Nearest Neighbors (KNN), Support Vector Machines (SVM), or even ensemble methods like Random Forests or Gradient Boosting).
 - **Training Pipeline:** Develop a clear pipeline that incorporates your chosen preprocessing, feature engineering/selection, and classification steps.
 - **Validation:** Implement a robust validation strategy (e.g. hold-out, k-fold cross-validation) to evaluate model performance and tune hyperparameters reliably. Avoid testing on the left-out subject during this phase.

- **Performance Metrics:** Select and calculate appropriate performance metrics (e.g., accuracy, precision, recall, F1-score, confusion matrix) to compare the models effectively.
- **Comments:** In this part you will be using Python (importing .csv files generated by MATLAB), as we have done throughout the course.

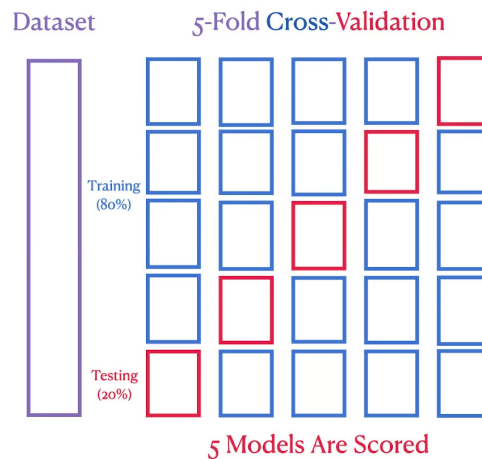


Figure 5: 5-fold cross-validation scheme. Taken from [a datacamp.com tutorial](https://datacamp.com/tutorial)

Model selection will also depend on your feature selection/extraction strategy. For example, if you use a *wrapping* approach and change the model, you may need to run feature selection for each model. If you choose a *filtering* approach, feature selection is model-independent, but less optimised for the chosen model. Keep in mind that, in this project, there is no “best” or “correct” way of choosing a validation strategy. You are free to try any alternative:

- All but one subject to train, **leaving one out** to test each time (selecting the best model as the one with best average performance),
- Implement a **hold-out** strategy by training with some subjects and testing with the rest,
- Training a model on each subject, leaving out some data to test each time (**stratified hold-out**), and then choosing the model also based on average performance.
- **Or something completely different** that you may come up with (perhaps devising an original approach, surprise your instructors!). The primary consideration here is to choose a strategy, interpret and understand the results obtained, and propose changes or improvements depending on them.

Also, take into account that we are dealing with a **3-class classification problem**, but it can be divided into **two 2-class classification problems**: rest vs motor imagery and then left vs right. In this case, discuss the suitability of each metric for this potentially imbalanced problem.

2.5 Phase 4: Final Model Assessment and Reporting

- **Objective:** Evaluate your best-performing model(s) on unseen data and prepare your final report and presentation.
- **Tasks:**
 - **Offline Testing:** Apply your final, trained model pipeline(s) to a separate, unseen test dataset (the left-out subject or your own recordings using the same protocol and OpenBCI headset). Report the performance on this dataset.

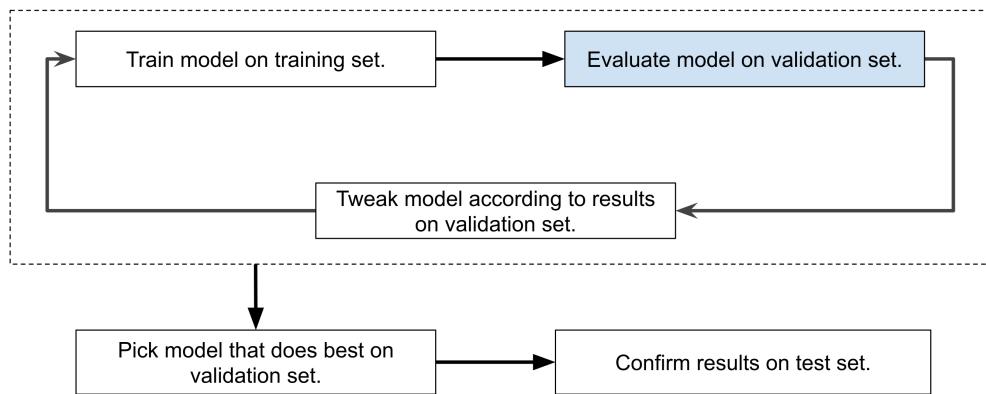


Figure 6: A good workflow for development and testing. Taken from [a machine learning course by Google](#).

- **Analysis and Discussion:** Analyse your results, discuss the strengths and weaknesses of your approach, compare the performance of different features and models, and reflect on challenges encountered. Include in the discussion a evaluation of your pipeline in terms of procesing load and response time (both training and applying the model), commenting on its suitability for real-time use. Does anything need to change? How would you make it work on real-time EEG signals?
 - **Real-Time Suitability:** Calculate your pipeline's algorithmic latency ($Total\ Latency = t_{window} + t_{preprocessing} + t_{inference}$) and comment on its suitability for a closed-loop BCI system.
 - **Final Report and Presentation:** Prepare a comprehensive report detailing your entire workflow, methodology, results, and conclusions. Deliver a group presentation summarising your project.
- **Comments:** More details on the deliverables in [Section 3](#).

3 Deliverables

3.1 Outputs of the Project

This project is expected to produce three outputs:

- Well-documented code implementing the full analysis pipeline. Predictably, MATLAB and Python code (either code files or Jupyter Notebooks).
- A final written report.
- A group presentation summarising the project and findings.

These three items will be included in 2 deliverables, described in [Section 3.2](#) and [Section 3.3](#).

3.2 Final Report

3.2.1 Contents

Your final report should be a clear, concise, and well-structured document detailing your data analysis project. Please ensure it includes the following sections:

- **Introduction:**
 - Clearly state the problem or research question being addressed.
 - Explain the motivation and importance of the analysis.
 - Define the specific objectives of your project.
 - Briefly outline the report's structure.
- **Data Description:**
 - Identify the data source(s).
 - Provide an overview of the data (variables, size, structure).
 - Detail all data preprocessing steps (cleaning, handling missing values, transformations, feature engineering) and justify why they were necessary.
- **Methodology:**
 - Describe the analytical techniques and models used (e.g., EDA, statistical tests, classification).
 - Justify your choice of methods based on the project objectives and data characteristics.
 - Mention key software or libraries used.
- **Results:**
 - Present your main findings objectively.
 - **Crucially:** Include **meaningful figures and tables** that are clearly labelled (titles, axes, legends) and have descriptive captions explaining what they show.
 - Reference all figures and tables within the text (e.g., "Figure 1 shows...").
 - Provide **accompanying comments** that explain the key insights derived from each figure and table. Do not just present visuals without explanation.
- **Discussion:**
 - **Interpret** the results – explain what they mean in the context of your problem/question.

- Relate your findings back to the initial objectives.
- Discuss any limitations of your data or methodology.
- Consider the implications or potential applications of your findings.

- **Conclusion:**

- Summarise the primary conclusions drawn from your analysis.
- Directly answer the initial research question based on your findings.
- Optionally, suggest potential areas for future work.

- **Ethical Use of AI (Transparency Section):**

- Briefly declare which Generative AI tools (e.g., ChatGPT, Gemini, Copilot) were used during the project and what they were used for (e.g., code debugging, text refining).
- If AI was used extensively, include a brief 1-2 sentence declaration here and direct the reader to a dedicated Annexe for a detailed log or explanation.

- **References:**

- List all cited sources (datasets, articles, tools) using a consistent citation style.

- **Annexes:**

- Include supplementary material (e.g., code snippets, extra plots) if it supports the main report but is too detailed for the body.
- **Do not include code as screenshots or images, code must be text that can be copied and pasted.** The use of $\LaTeX$ is highly recommended. You may use this document as a reference if you are unsure where to start; the $\LaTeX$ source is available [here](#).



Figure 7: “To the Dark Side, a Jedi turns, hmmm, each time a code screenshot to a document you paste, yes.” — Master Yoda

Overall Expectation: The report should demonstrate a clear understanding of the problem, thoughtful application of data analysis techniques, and the ability to derive and communicate meaningful conclusions supported by well-presented evidence (figures, tables, and insightful commentary).

3.2.2 Delivery

- The report is expected to be delivered [via ATENEA](#) **before the end of 7th June**.
- The document should be **one PDF document**. Additional files or information that cannot be included as annexes (such as .csv files or .ipynb notebooks), will be added in an **additional .zip file**. The **maximum allowed size** for each file is **200MB**.

3.3 Presentation

3.3.1 Contents

Your presentation, similarly to the report, should be a clear, concise, and well-structured document summarising your project. It can have a similar structure to the report, but it is not compulsory to follow it.

3.3.2 Delivery

- The presentation document is expected to be delivered [via ATENEA](#) **before the end of 7th June**.
- The document should be **one PDF document**. No additional files will be accepted. The **maximum allowed size** for the file is **200MB**.
- Each group of students will have a slot of 10 min to 15 min to present their projects **on 8th June, just after the final exam**.
- You may use a **PDF or Powerpoint** file and the computer in the room for presenting (recommended), or your own laptop if you need different software.

3.4 Grading

The final score of the project will be a weighted average:

- 70 % corresponds to the delivered documents (Project PDF, Presentation PDF).
- 30 % corresponds to the group presentation.

Check all the criteria used to assess the project in [Annexe B](#), and to assess the presentation in [Annexe C](#).

4 Collaboration and Expectations

This project relies heavily on effective teamwork and collaborative effort from everyone involved. To foster a positive and productive working environment, please ensure your group prioritises the following behaviours:

- **Clear and open communication:** Actively listen to each other and express ideas respectfully, ensuring everyone's voice can be heard.
- **Equitable task distribution:** Share responsibilities fairly amongst all team members, considering individual strengths and development goals where appropriate. Agree on roles and tasks collectively.
- **Regular team meetings:** Schedule and attend regular check-ins to discuss progress, address any challenges, and maintain alignment within the group organisation.

Please bear in mind that:

- All individuals in the group are expected to make significant contributions throughout every phase of the project. Active participation from everyone is essential for success.
- We strongly encourage you to explore different avenues, experiment with techniques, and apply creative thinking to your approach, whilst always basing your technical decisions on sound data analysis principles.



Figure 8: Generated using [Gemini 2.5 Pro](#). Students must report and cite any AI tools used for content generation or code debugging as per UPC guidelines.

5 Acknowledgements

The data in this project has been acquired and curated by Eng. Jan Pinyol, under the supervision of Prof. Dr. Sergio Romero, and with help from MSc. Eng Oziel R. Cantú. Jan and Sergio have also contributed several figures and code examples. Profs. Dr. Alejandro Bachiller, Dr. Mónica Rojas, and Dr. Sergio Romero have also contributed with their experience in BCI and teaching in the HMI course, as well as reviewing this document. I also want to extend my gratitude to the volunteers at the [BIOART Group](#) for their selfless contribution to our EEG recording study. Thanks to you all!

Joan F. Alonso
Barcelona, April 2026

Annexe A Description of the Database

The EEG data for this project were acquired using a 16-channel OpenBCI system, with the EEG signals referenced to the right mastoid. EEGLab was employed solely to read the data files generated by LabRecorder, allowing for the alignment of EEG data with temporal markers from the Psychopy experiment. No additional common referencing was performed in EEGLab (although the .mat files may indicate the contrary by default).

A.1 Channel Information

- The channel labels in the .mat data files contain the raw output from the OpenBCI system (see [Figure 9](#)).
- Data transmission was managed via a command-line interface, which also facilitated the configuration of data transmission parameters, including amplifier gain.
- The labels represent the following standard 10-20 system locations: F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, P7, P3, Pz, P4, and P8.
- A 16th channel is present due to amplifier configuration but must be excluded from the analysis.

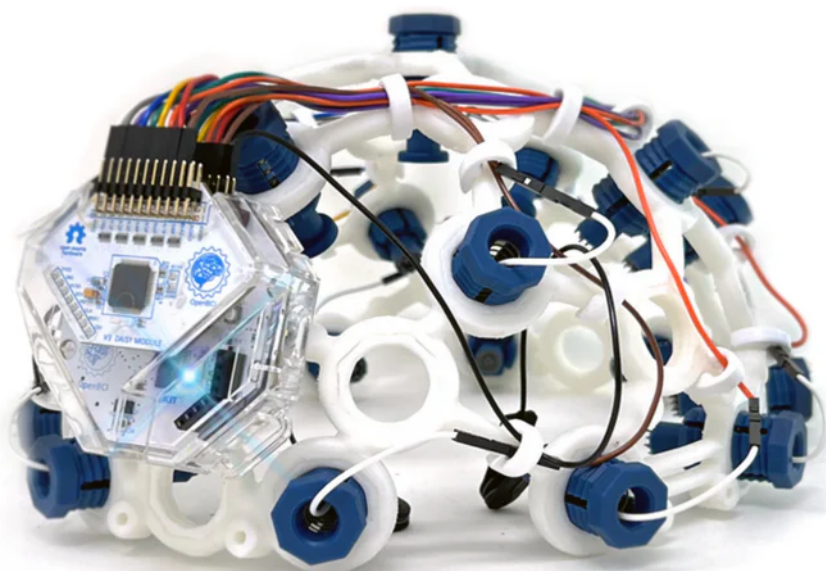


Figure 9: Ultracortex EEG headset. Taken from [the OpenBCI documentation](#).

A.2 Experimental Protocol

The experimental protocol involved 8 runs of 60 trials each (20 for each class), with breaks in between. The sequence of tasks was randomised within each run. Specifically, each run contained **20 trials of right-hand task**, **20 trials of left-hand task**, and **20 trials of rest**, which appeared in random order.

The first **two runs consisted of motor executions (ME)**, while the remaining **six runs involved motor imagery (MI)**. This resulted in a total of 480 trials (8 runs $\times$ 60 trials), distributed as 160 trials for each condition (right-hand, left-hand, rest).

Thus, for motor imagery (MI) classification, data start at the 121st trial. This subset comprises 360 MI trials (6 runs $\times$ 60 trials), with 120 trials per condition (right-hand, left-hand, rest).

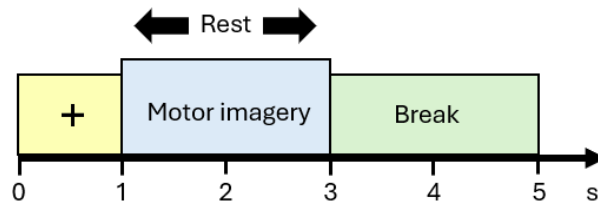


Figure 10: Paradigm used in the motor imagery trials. Image by J. Pinyol.

A.3 Trial Structure

The experiment was programmed using PsychoPy. Each trial had an approximate duration of 5 s and followed this structure (see [Figure 10](#)):

- **Fixation Cross:** 5 s + random duration in the range of ± 0.2 s.
- **Task Period:** 2 s. Here the volunteer was shown either an arrow indicating right or left (for motor imagery), or the word *Rest*.
- **Inter-Trial Interval:** 2 s + random duration in the range of ± 0.2 s.

Annexe B Project Report Rubric

Table 1: Project Report Rubric

Aspect to Evaluate	Level 4 Outstanding	Level 3 Commendable	Level 2 Satisfactory	Level 1 Insufficient	Level 0 No Evidence	Weight
1 Introduction and Problem Description	Provides a comprehensive and insightful introduction, clearly defining the problem, strong motivation, precise objectives, and a clear report outline. Context is exceptionally well-established.	Provides a clear introduction, defining the problem, motivation, objectives, and report outline. Context is well-established.	Provides a basic introduction, outlining the problem, motivation, objectives, and report structure. Some elements may lack detail or clarity. Context may be partially missing.	Introduction is unclear, incomplete, or significantly flawed. Problem, motivation, objectives, or structure are poorly defined or missing.	No introduction or problem description provided.	10%
2 Structure, Format, and Language	Exceptionally well-organised report following all required sections. Professional formatting, excellent figures/tables integration. Language is precise, clear, concise, and error-free academic English.	Well-organised report adhering to structure. Good formatting and integration of visuals. Language is clear, mostly concise, and uses appropriate academic English with minimal errors.	Generally organised report, most sections present. Formatting is adequate but may have inconsistencies. Language is understandable but may lack conciseness or contain several errors.	Poorly organised, sections missing or illogical. Formatting is inconsistent or unprofessional. Language is unclear, verbose, or contains significant errors.	Report structure/format makes it unreadable or unevaluable.	15%
3 Methodology and Tool Usage (MATLAB, Python)	Detailed, robust, and expertly justified methodology. Excellent choice and application of algorithms/tools. Demonstrates deep understanding of techniques used.	Clear and well-justified methodology. Good choice and correct application of algorithms/tools. Demonstrates solid understanding.	Basic description of methodology. Adequate choice of algorithms/tools, but justification or application may have minor flaws. Demonstrates basic understanding.	Methodology is inadequate, unclear, or poorly justified. Incorrect choice or application of algorithms/tools. Lacks understanding of techniques.	No methodology described or tools used completely inappropriately.	15%
4 Results Presentation	Presents highly insightful results using highly relevant metrics. Figures/tables are exceptionally clear, well-labelled, informative, perfectly integrated, and expertly explained in the text.	Presents clear results using appropriate metrics. Figures/tables are clear, well-labelled, adequately integrated, and well-explained in the text.	Presents basic results, some relevant metrics might be missing. Figures/tables are present but may lack clarity, labels, or sufficient explanation/integration in the text.	Results are unclear, irrelevant, or poorly presented. Metrics are inappropriate or missing. Figures/tables are confusing, poorly labelled, or absent.	No results presented.	15%
5 Discussion and Critical Interpretation	Provides insightful and deep interpretation of results, critically analysing strengths/weaknesses. Effectively relates findings to objectives and BCI context. Discusses implications and limitations thoroughly.	Provides clear interpretation of results, analysing strengths/weaknesses. Relates findings to objectives and context well. Discusses implications and limitations adequately.	Provides basic interpretation of results. Some analysis of strengths/weaknesses present. Connection to objectives/context is made but may be superficial. Basic discussion of limitations.	Interpretation is minimal, incorrect, or superficial. Little to no analysis of strengths/weaknesses or connection to objectives/context.	No discussion or interpretation provided.	15%
6 Conclusions and Originality	Draws strong, well-supported conclusions directly addressing all objectives. Answers the research question definitively. Demonstrates significant original thought or creative approaches.	Draws clear conclusions addressing main objectives. Answers the research question. Shows some evidence of original thought or creative problem-solving.	Draws basic conclusions, but they may not fully address all objectives or the research question. Lacks originality, strictly following standard procedures.	Conclusions are weak, unsupported by results, or do not address objectives. Answers are vague or incorrect.	No conclusions provided.	15%

Table 1: Project Report Rubric (continued)

Aspect to Evaluate	Level 4 Outstanding	Level 3 Commendable	Level 2 Satisfactory	Level 1 Insufficient	Level 0 No Evidence	Weight
7 Code Quality and Reproducibility	Code is flawless, excellently structured, highly readable, and thoroughly commented. It runs perfectly and reproduces all results effortlessly. Best coding practices are followed.	Code is clear, reasonably structured, and adequately documented. It runs correctly and reproduces the main results. Good practices are generally followed.	Code is generally understandable but may lack structure or sufficient documentation. It runs but may require minor fixes or clarifications to reproduce results. Basic coding practices followed.	Code is difficult to understand, poorly structured, or lacks documentation. It fails to run correctly or reproduce results without significant effort.	No code submitted, or code is completely non-functional.	15%

Annexe C Presentation Rubric

Table 2: Presentation Rubric

Aspect to Evaluate	Level 4 Outstanding	Level 3 Commendable	Level 2 Satisfactory	Level 1 Insufficient	Level 0 No Evidence	Weight
1 Organisation and Structure	Exceptionally clear and logical progression. Introduction captures attention; conclusion is strong and memorable. Transitions are seamless.	Clear and logical structure. Good introduction and conclusion. Smooth transitions between topics.	Adequate structure, though the flow may occasionally be disjointed. Introduction and conclusion are present but basic.	Poorly organised, making it difficult to follow. Weak or missing introduction/conclusion. Abrupt transitions.	No discernible structure.	15%
2 Content and Depth	Demonstrates a deep mastery of the subject matter. Content is highly relevant, accurate, and perfectly tailored to the audience. Complex concepts explained simply.	Demonstrates a good understanding. Content is relevant and accurate. Most concepts are explained clearly.	Demonstrates basic understanding. Content is mostly relevant, but some explanations may lack depth or clarity.	Demonstrates poor understanding. Content is inaccurate, irrelevant, or lacks sufficient depth.	Content is completely irrelevant or incorrect.	25%
3 Visuals and Delivery	Visual aids are highly professional, informative, and perfectly support the talk. Delivery is engaging, clear, well-paced, and demonstrates confidence. Excellent use of language.	Visual aids are clear, relevant, and effectively used. Delivery is clear, well-paced, and confident. Good use of language.	Visual aids are adequate but may have minor issues (e.g., clutter, relevance). Delivery is generally clear but may lack engagement or have minor flaws (pace, clarity).	Visual aids are poor quality, confusing, or used ineffectively. Delivery is unclear, hesitant, poorly paced, or lacks engagement.	No visual aids used (if required) or delivery is unintelligible.	20%
4 Teamwork and Time Mngmt.	Perfect coordination, smooth transitions, and balanced participation among group members. Presentation adheres well to the time limit.	Adequate coordination and participation, though some imbalance or minor issues with transitions may exist. Generally adheres to time limit, may be slightly over/under.	Poor coordination, awkward transitions, or significantly unbalanced participation. Poor adherence to time limits (significantly over/under).	No evidence of teamwork, or time management hinders presentation.	Only one speaker.	10%
5 Answering Questions	Demonstrates mastery by providing exceptionally clear, accurate, concise, and insightful answers to all questions. Handles challenging questions effectively.	Provides clear, accurate, and thoughtful answers to questions, demonstrating a solid understanding of the project. Handles most questions well.	Provides generally adequate answers, but some may lack clarity, depth, or accuracy. Demonstrates basic understanding but struggles with complex questions.	Unable to answer most questions accurately or clearly. Responses are confused, incorrect, or demonstrate a lack of understanding.	Does not attempt to answer questions or responses are irrelevant.	30%